Directed Search with Rational Inattention

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November 3, 2026

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Preliminary draft

Abstract. This paper develops a directed search model for estimating workers' job preferences and information costs from application data. Workers observe firm types but not realized wages and choose what wage information to acquire before deciding where to apply. Information costs take the Shannon entropy form and can vary across worker types. I define a rationally inattentive search equilibrium in which application choices determine congestion across submarkets, while congestion shapes job-finding probabilities and the returns to information. I establish equilibrium existence when the wage distribution has finite support and job-finding probabilities depend continuously on workers' application probabilities. The proof's key insight is to use the linear-fractional structure of the generalized logit formula to establish existence without requiring a unique solution to workers' information-acquisition problems. I also establish conditions for identifying worker-type-specific information costs using twin job types with identical amenity values but different wages.

Keywords. Directed search; matching; rational inattention; information costs.

JEL classification. D83, J64, C57, D82.

1. Introduction

Learning about jobs takes time and effort. Many vacancies do not advertise wages (Banfi and Villena-Roldán, 2019; Marinescu and Wolthoff, 2020). Even when information is provided, workers do not always pay close attention to it (Mas and Pallais, 2017). Workers can investigate jobs more closely before applying, but the value of doing so must be weighed against the cost. Rational inattention models this decision by allowing workers to choose what to learn, balancing the expected gains from better decisions against information costs (Sims, 2003).

Estimating workers' preferences from job applications therefore requires accounting for their chances of being hired and the cost of learning about available jobs. A worker may apply to a lower-wage job because she values its amenities, because it offers a better chance of being hired, or because she has learned little about the alternatives. If the cost of learning differs across workers, differences in their responses to wages need not reflect differences in preferences alone. Accounting for these costs also matters for evaluating wage disclosure, job-search assistance, and policies that change wages, because workers' responses depend on what they learn before applying.

This paper develops a directed search model with heterogeneous workers and firms as a framework for structural estimation with costly information acquisition. The researcher observes wages and application shares and seeks to recover workers' information costs and valuations of non-wage amenities. Amenities depend on both worker and firm type, while information costs can differ across worker types. I establish equilibrium existence, derive identification conditions, and formulate estimation approaches for these parameters.

Workers observe firm types but not realized wages. They share a probability distribution over possible wages and acquire information before choosing where to apply. Information costs are proportional to the reduction in Shannon entropy. Under this specification, Matějka and McKay (2015) derive a generalized logit formula, which I use to describe workers' application probabilities. Workers apply to submarkets, where the matching technology maps the application probabilities of different worker types into job-finding probabilities. I define a rationally inattentive search equilibrium (RISE) by requiring workers' information and application choices to be optimal at these job-finding probabilities. This consistency condition matters for estimation: a change in information costs affects where workers apply and therefore changes the competition for jobs faced by other workers.

I prove that an equilibrium exists when the wage distribution has finite support and job-finding probabilities depend continuously on application probabilities. The result allows workers to have several optimal information-acquisition strategies. The proof exploits the linear-fractional structure of the generalized logit formula: *a linear-fractional map with a positive denominator maps a convex set to a convex set* (Boyd and Vandenberghe, 2004, Section 2.3.3).

Together with workers' optimality conditions, this property supplies the convexity needed to apply Kakutani's fixed-point theorem. At the resulting fixed point, optimal applications reproduce the congestion to which workers respond.

I also establish conditions for recovering worker-type-specific information costs from conditional shares, which describe applications at a particular wage vector, and ex-ante shares, which average over wage realizations. The comparison uses *twin* job types with the same amenity value for a given worker type but different wages. When the twins have equal job-finding probabilities, their common amenity value cancels from the difference in expected payoffs. Given known matching technology, the twins' difference in expected wage income and their choice shares yield a closed-form expression for the information cost. The restriction can be applied separately by worker type, allowing information costs to be recovered without imposing a common valuation of amenities across workers. With a zero-payoff reference option, I also give conditions for jointly identifying the information cost and the twins' common amenity value even when their job-finding probabilities differ.

I outline likelihood-based estimators of amenity values and information costs using either estimated ex-ante shares or joint optimization over parameters and shares. With job-finding probabilities taken as given, these estimators adapt conditional-choice-probability and constrained-optimization methods (Hotz and Miller, 1993; Aguirregabiria and Mira, 2002; Su and Judd, 2012). Full-equilibrium estimation also imposes the matching equations.

Endogenous learning also changes the choice patterns a model can explain. Adding a job can increase the probability of applying to an existing job by changing what workers learn. Rational inattention also permits exact zero shares and kinks in choice probabilities (Matějka and McKay, 2015; Caplin et al., 2019; Liao, 2024). To motivate the role of learning, I compare occupational choices using vacancy-viewing and saving records from the platform studied by Belot et al. (2022). Occupational choices differ across workers who view different sets of occupations. Because workers choose which vacancies to view, this comparison is descriptive; it does not isolate the effect of information acquisition.

Related literature

The paper builds on search and matching models that explain how frictions affect the allocation of workers to jobs (Diamond, 1971; Mortensen, 1982; Pissarides, 1985). Directed search makes workers' application choices and congestion jointly determined (Peters, 1984, 1991; Moen, 1997; Peters, 2010), and heterogeneous-agent models use this structure to study sorting across jobs (Shi, 2001; Eeckhout and Kircher, 2010). Cheremukhin et al. (2020) model costly targeting of potential partners and study its implications for recovering preferences. In the present model, workers acquire information about wages before directing their applications to firm types. The identification analysis uses twin jobs to recover information costs from application shares while allowing amenity values and information costs to vary across worker types.

The information problem follows Sims (2003). Under Shannon costs, Matějka and McKay (2015) derive a generalized logit formula for conditional choice probabilities, connecting rational inattention to discrete-choice models (Luce, 1959; McFadden, 1973). The ex-ante weights in this formula are chosen optimally and may be zero, a feature underlying the consideration sets studied by Caplin et al. (2019). In a search market, workers' choices also affect the returns to information through congestion.

Caplin and Dean (2015), Caplin et al. (2022), and Dean and Neligh (2023) study the behavioral restrictions and empirical content of information-acquisition models. Liao (2024) develops identification results for rational-inattention discrete choice. In the present paper, the identifying restrictions come from job search: equal amenities and equal job-finding probabilities isolate wage variation within a pair of twin jobs. The resulting comparison identifies the information cost without requiring the twins' common amenity value to be known.

Section 2 presents the model and the existence result. Section 3 develops identification, and Section 4 discusses estimation. Section 5 compares the model's choice implications with familiar benchmarks and discusses the empirical choice patterns. Section 6 concludes. The appendix contains notation, the existence proof, and an illustrative matching example.

2. The Model

There is a continuum of workers of mass one and a continuum of firms of mass $m_{\mathcal{J}} > 0$. Workers and firms have observable types $x \in \mathcal{X} = \{1, \ldots, |\mathcal{X}|\}$ and $y \in \mathcal{Y} = \{1, \ldots, |\mathcal{Y}|\}$, respectively. Both type sets are finite. Matching is one-to-one, and agents are indifferent between prospective partners of the same type. A type-x worker employed by a type-y firm receives utility $\alpha(x, y) + w(y)$, where $\alpha(x, y)$ is the value of non-wage amenities and $w(y)$ is the wage. Utility when unmatched is normalized to zero, so amenities are measured relative to that outcome.

The wage vector $\boldsymbol{w} = (w(1), \ldots, w(|\mathcal{Y}|))$ is drawn from a distribution G. A realization of this vector is a *state*. Workers observe firm types and share the prior G, but learn about realized wages through costly information acquisition. After receiving information, each worker chooses one job type and enters its submarket. Within a submarket, a matching technology determines which workers obtain jobs, as in Moen (1997) and Eeckhout and Kircher (2010). Wages are assumed to have finite first moments.

Let $\rho_y^x(\boldsymbol{w})$ be the probability that a type-x worker is hired after applying to submarket y in state $\boldsymbol{w}$. Her expected payoff from applying is

$$v_y^x(\boldsymbol{w}) = \rho_y^x(\boldsymbol{w})\big(\alpha(x, y) + w(y)\big). \tag{2.1}$$

Job-finding probabilities depend on the applications of other workers. A higher wage can therefore increase both the return to employment and the competition for a job. Section 2.2 specifies how applications determine job-finding probabilities.

2.1. Workers' information and application choices

Consider a worker of type x, taking the functions ρ_y^x as given. I suppress the type index in this subsection. A signal provides information about the wage vector. An information structure specifies the distribution of that signal in each wage state; formally, it is a joint distribution $F(\boldsymbol{s}, \boldsymbol{w})$ over signals and wage states whose wage marginal is G. After observing a signal $\boldsymbol{s}$, the worker forms the posterior $F(\cdot \mid \boldsymbol{s})$ and chooses a job type to maximize her posterior expected payoff. For a posterior B, define

$$V(B) = \max_{y \in \mathcal{Y}} \int v_y(\boldsymbol{w})\, dB(\boldsymbol{w}).$$

The worker chooses an information structure before observing the signal. After the wage state and signal are realized, she updates her beliefs, applies to a submarket, and participates in matching. Her information choice maximizes expected payoff net of information costs:

$$\max_{F : F_{\boldsymbol{w}} = G} \int \int V\big(F(\cdot \mid \boldsymbol{s})\big)\, dF(\boldsymbol{s} \mid \boldsymbol{w})\, dG(\boldsymbol{w}) - \widehat{c}(F). \tag{2.2}$$

Following Matějka and McKay (2015), the information cost is

$$\widehat{c}(F) = k\big[H(G) - \mathbb{E}_{\boldsymbol{s}} H\big(F(\cdot \mid \boldsymbol{s})\big)\big], \qquad k > 0, \tag{2.3}$$

where H denotes Shannon entropy. For a distribution with finite support and probabilities $(p_j)_j$, $H = -\sum_j p_j \log p_j$, with $0 \log 0 = 0$. When G has finite support, the bracketed term is the mutual information between signals and wages. More generally, the information cost is $kD_{\mathrm{KL}}(F \| F_{\boldsymbol{s}} \otimes G)$, where $F_{\boldsymbol{s}}$ is the signal marginal and D_{KL} is relative entropy. A signal that leaves the prior unchanged has zero information cost. A worker can therefore choose to acquire no information and base her application on the prior.

Definition 2.1 (Information and action strategy)**.** An optimal strategy is a pair (F, a) such that F solves (2.2) and $a(B) = (a_y(B))_{y \in \mathcal{Y}}$ is a probability distribution over actions satisfying

$$a_y(B) > 0 \quad \Longrightarrow \quad y \in \arg\max_{j \in \mathcal{Y}} \mathbb{E}_B[v_j]$$

for each posterior B induced by F. The action rule may randomize among jobs with the same posterior expected payoff.

The information problem has an equivalent formulation in terms of conditional application probabilities (Matějka and McKay, 2015; Denti et al., 2020). Signals leading to the same application can be grouped into a recommendation to choose that job type without changing expected payoffs or increasing information costs. The worker therefore needs at most $|\mathcal{Y}|$ recommendations. This representation is also used by Ravid (2020) and Cusumano et al. (2024).

Definition 2.2 (Recommendation strategy)**.** A recommendation strategy is a collection of measurable functions $\boldsymbol{\beta} = (\beta_y)_{y\in\mathcal{Y}}$, where $\beta_y : \operatorname{supp}(G) \to [0,1]$ gives the probability of an application to job type y conditional on the wage vector. The probabilities satisfy $\sum_y \beta_y(\boldsymbol{w}) = 1$ at every wage state. Its ex-ante choice probabilities are $\bar{\beta}_y = \int \beta_y(\boldsymbol{w})\, dG(\boldsymbol{w})$.

Proposition 2.1 (Dual formulation of the information problem; Matějka and McKay, 2015)**.** *A recommendation strategy is induced by an optimal information and action strategy if and only if it solves*

$$\max_{\boldsymbol{\beta}} \int \sum_{y\in\mathcal{Y}} v_y(\boldsymbol{w})\beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}) - c(\boldsymbol{\beta}, G), \tag{2.4}$$

where

$$c(\boldsymbol{\beta}, G) = k\left[\int \sum_{y\in\mathcal{Y}} \beta_y(\boldsymbol{w}) \log \beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}) - \sum_{y\in\mathcal{Y}} \bar{\beta}_y \log \bar{\beta}_y\right]. \tag{2.5}$$

For $\bar{\beta}_y > 0$, recommendation y induces the posterior $dB_y(\boldsymbol{w}) = \beta_y(\boldsymbol{w})\, dG(\boldsymbol{w})/\bar{\beta}_y$. At an optimum,

$$\int \big[v_y(\boldsymbol{w}) - v_j(\boldsymbol{w})\big]\beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}) \geq 0 \qquad \text{for every } j \in \mathcal{Y},$$

so following the recommendation is optimal. Otherwise, redirecting recommendation y to a better action would increase expected payoff without increasing mutual information.

Figure 1 summarizes the two formulations. The first chooses an information structure and then an action after each signal; the second chooses the probability of each action conditional on the wage state.

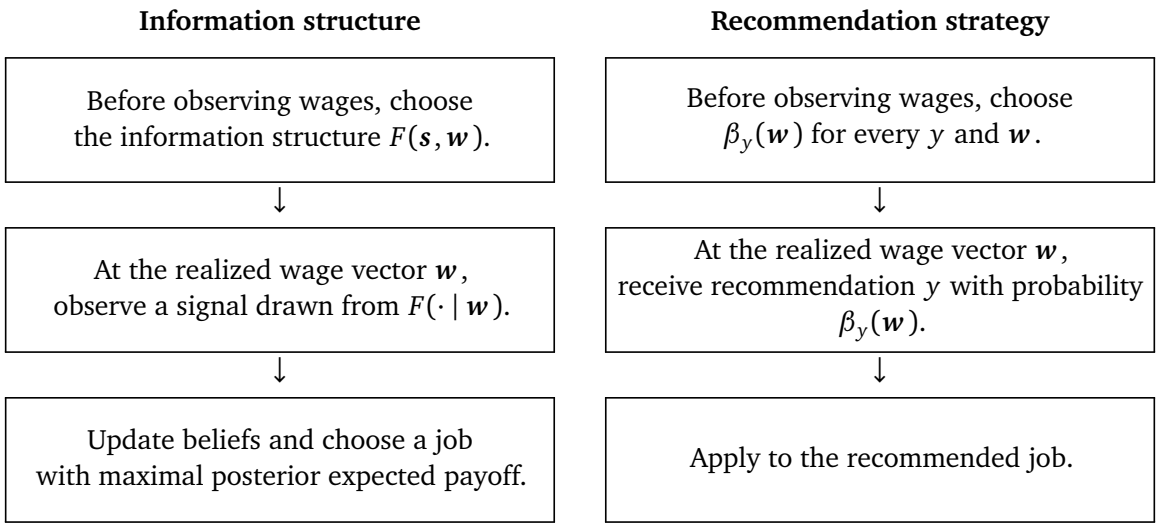


Figure 1: Information structures and recommendation strategies.

Let $\Delta(\mathcal{Y}) = \{q \in \mathbb{R}_+^{|\mathcal{Y}|} : \sum_y q_y = 1\}$ be the probability simplex. Introduce an auxiliary vector $q \in \Delta(\mathcal{Y})$. For any recommendation strategy,

$$k \int D_{\mathrm{KL}}\big(\boldsymbol{\beta}(\boldsymbol{w}) \| q\big)\, dG(\boldsymbol{w}) = c(\boldsymbol{\beta}, G) + k D_{\mathrm{KL}}\big(\bar{\boldsymbol{\beta}} \| q\big).$$

Minimizing over q gives $q = \bar{\boldsymbol{\beta}}$ and recovers the original information cost. Thus the worker can optimize jointly over q and $\boldsymbol{\beta}$; holding q fixed gives a separate problem at each wage state:

$$\max_{p \in \Delta(\mathcal{Y})} \left\{ \sum_y p_y v_y(\boldsymbol{w}) - k \sum_y p_y \log \frac{p_y}{q_y} \right\}. \tag{2.6}$$

The second term is k times the Kullback–Leibler divergence from q. If $q_y = 0$, finite information cost requires $p_y = 0$.

Proposition 2.2 (Matějka and McKay (2015))**.** *For $k > 0$, an optimal recommendation strategy satisfies*

$$\beta_y(\boldsymbol{w}) = \frac{\bar{\beta}_y \exp\{v_y(\boldsymbol{w})/k\}}{\sum_{j\in\mathcal{Y}} \bar{\beta}_j \exp\{v_j(\boldsymbol{w})/k\}}, \qquad y \in \mathcal{Y}, \tag{2.7}$$

where the ex-ante weights solve

$$\bar{\boldsymbol{\beta}} \in \arg\max_{q\in\Delta(\mathcal{Y})} \int k \log\left(\sum_{y\in\mathcal{Y}} q_y \exp\{v_y(\boldsymbol{w})/k\}\right) dG(\boldsymbol{w}). \tag{2.8}$$

Conversely, optimal weights in (2.8)*, together with* (2.7)*, generate an optimal recommendation strategy.*

The conditional probabilities have a logit form, with weights determined by the information problem. Each weight is the probability of applying to that job type, averaged over wage realizations. Appendix B verifies that the probabilities generated by optimal weights average back to those weights, including when some weights are zero.

2.2. Rationally inattentive search equilibrium

In equilibrium, job-finding probabilities depend on the distribution of applications. For each worker type x and job type y, let $r_y^x : [0,1]^{|\mathcal{X}|} \to [0,1]$ be a job-finding probability function. In state $\boldsymbol{w}$,

$$\rho_y^x(\boldsymbol{w}) = r_y^x\left(\beta_y^1(\boldsymbol{w}), \ldots, \beta_y^{|\mathcal{X}|}(\boldsymbol{w})\right). \tag{2.9}$$

The function incorporates the masses of workers and vacancies and the allocation rule within the submarket. Each individual worker has zero mass and takes the application profile of the population as given.

Definition 2.3 (Rationally inattentive search equilibrium)**.** A rationally inattentive search equilibrium (RISE) is a profile $\boldsymbol{\beta}^* = (\beta_y^{*x})_{x\in\mathcal{X},y\in\mathcal{Y}}$ satisfying the following conditions for every worker type x and every wage state in the support of G.

(i) Conditional application probabilities satisfy

$$\beta_y^{*x}(\boldsymbol{w}) = \frac{\bar{\beta}_y^{*x}\exp\{v_y^{*x}(\boldsymbol{w})/k_x\}}{\sum_{j\in\mathcal{Y}}\bar{\beta}_j^{*x}\exp\{v_j^{*x}(\boldsymbol{w})/k_x\}}, \tag{2.10}$$

where $\bar{\beta}_y^{*x} = \int \beta_y^{*x}(\boldsymbol{w})\,dG(\boldsymbol{w})$ and $v_y^{*x}(\boldsymbol{w}) = \rho_y^{*x}(\boldsymbol{w})(\alpha(x,y)+w(y))$.

(ii) Ex-ante application probabilities solve

$$\bar{\boldsymbol{\beta}}^{*x} \in \arg\max_{q\in\Delta(\mathcal{Y})} \int k_x \log\left(\sum_{y\in\mathcal{Y}} q_y \exp\{v_y^{*x}(\boldsymbol{w})/k_x\}\right) dG(\boldsymbol{w}). \tag{2.11}$$

(iii) Job-finding probabilities are generated by the application profile:

$$\rho_y^{*x}(\boldsymbol{w}) = r_y^x\big(\beta_y^{*1}(\boldsymbol{w}),\ldots,\beta_y^{*|\mathcal{X}|}(\boldsymbol{w})\big). \tag{2.12}$$

Conditions (i) and (ii) give optimal information acquisition and applications at the prevailing job-finding probabilities. Condition (iii) requires those probabilities to agree with the applications generated by workers' choices.

Proposition 2.3 (Existence of RISE)**.** *Suppose $k_x > 0$ for every worker type x, the wage distribution G has finite support, and each job-finding function r_y^x is continuous in the application probabilities of all worker types. Then a RISE exists.*

The novelty of the proof lies in using the linear-fractional structure of the generalized logit formula to establish equilibrium existence. Specifically, *a linear-fractional function with a positive denominator maps a convex set to a convex set* (Boyd and Vandenberghe, 2004, Section 2.3.3, p. 42). Together with workers' optimality conditions, this property yields the convexity required for Kakutani's fixed-point theorem, even when optimal ex-ante weights are nonunique.

Let $K = \prod_{x\in\mathcal{X},\,\boldsymbol{w}\in\mathrm{supp}(G)}\Delta(\mathcal{Y})$ be the set of application profiles. A conjectured profile $\boldsymbol{b}\in K$ induces the payoffs

$$v_y^x(\boldsymbol{w};\boldsymbol{b}) = r_y^x\big((b_y^{x'}(\boldsymbol{w}))_{x'\in\mathcal{X}}\big)\big[\alpha(x,y)+w(y)\big].$$

Let $B^x(\boldsymbol{b})$ be the maximizers in (2.11) at these payoffs, holding $\boldsymbol{b}$ fixed while optimizing over q. Define

$$Z_{\boldsymbol{w}}^x(q;\boldsymbol{b}) = \sum_j q_j \exp\{v_j^x(\boldsymbol{w};\boldsymbol{b})/k_x\}, \qquad T_y^x(\boldsymbol{w};q,\boldsymbol{b}) = \frac{q_y\exp\{v_y^x(\boldsymbol{w};\boldsymbol{b})/k_x\}}{Z_{\boldsymbol{w}}^x(q;\boldsymbol{b})}.$$

Let $H(\boldsymbol{b})$ collect the application profiles generated by optimal weight vectors at the conjectured profile $\boldsymbol{b}$. An equilibrium requires a fixed point of the correspondence $H : K \rightrightarrows K$:

$$\boldsymbol{\beta}^* \in H(\boldsymbol{\beta}^*), \qquad H(\boldsymbol{b}) = \left\{ \left(T_y^x(\boldsymbol{w}; q^x, \boldsymbol{b}) \right)_{x,y,\boldsymbol{w}} : q^x \in B^x(\boldsymbol{b}) \text{ for every } x \right\}.$$

Workers' optimal applications then reproduce the profile determining congestion.

Concavity of the outer objective makes $B^x(\boldsymbol{b})$ convex, and the logit map T^x has the linear-fractional form above. To apply the convexity-preserving property jointly across wage states, take $q, \tilde{q} \in B^x(\boldsymbol{b})$. Optimality and strict concavity of the logarithm give

$$Z_{\boldsymbol{w}}^x(q; \boldsymbol{b}) = Z_{\boldsymbol{w}}^x(\tilde{q}; \boldsymbol{b}) \qquad \text{for every } \boldsymbol{w} \in \operatorname{supp}(G).$$

Hence, for $\theta \in [0,1]$,

$$T_y^x\big(\boldsymbol{w}; \theta q + (1-\theta)\tilde{q}, \boldsymbol{b}\big) = \theta T_y^x(\boldsymbol{w}; q, \boldsymbol{b}) + (1-\theta) T_y^x(\boldsymbol{w}; \tilde{q}, \boldsymbol{b}) \qquad \text{for every } y, \boldsymbol{w}.$$

Thus $H(\boldsymbol{b})$ is convex jointly across wage states. The domain K is compact and convex; the maximum theorem and continuity of T give nonempty compact values and upper hemicontinuity of H. Kakutani's theorem therefore yields a fixed point. The associated optimal weights satisfy $q_y^x = \int \beta_y^{*x}(\boldsymbol{w})\, dG(\boldsymbol{w})$, so the fixed point is a RISE. The full proof is in Appendix B.

Appendix C provides the equivalent formulation directly in terms of recommendation strategies, and Appendix D illustrates the equilibrium conditions in a two-type matching market.

3. Identification

The identification question is whether application shares and wages determine the information cost k_x and amenity values $\alpha(x, y)$ for a given worker type x. The analysis uses two population shares:

$$\mu(y \mid x, \boldsymbol{w}) = \beta_y^{*x}(\boldsymbol{w}), \qquad m(y \mid x) = \bar{\beta}_y^{*x} = \int \mu(y \mid x, \boldsymbol{w})\, dG(\boldsymbol{w}).$$

The conditional share $\mu(y \mid x, \boldsymbol{w})$ describes applications at a given wage vector. The ex-ante share $m(y \mid x)$ averages those applications over wage realizations. I take both shares to be identified from the data and the job-finding functions r_y^x to be known. Evaluating these functions at the conditional shares then gives the job-finding probabilities $\rho_y^x(\boldsymbol{w})$.

3.1. Identification using twin jobs

Strictly positive shares allow comparisons of log choice odds.

Assumption 3.1 (Positive shares)**.** For every worker type x, job type y, and wage vector $\boldsymbol{w}$ in the support of G, $\mu(y \mid x, \boldsymbol{w}) > 0$.

Under this assumption, taking logs in (2.10) and comparing any two job types gives

$$\log \frac{\mu(y \mid x, \boldsymbol{w})}{\mu(y' \mid x, \boldsymbol{w})} - \log \frac{m(y \mid x)}{m(y' \mid x)} = \frac{v_y^x(\boldsymbol{w}) - v_{y'}^x(\boldsymbol{w})}{k_x}. \tag{3.1}$$

The left-hand side compares the two jobs' log choice odds at a given wage vector with their ex-ante log choice odds. The right-hand side is the expected application payoff difference in units of the information cost.

For worker type x, job types y and y' are *twins* if $\alpha(x, y) = \alpha(x, y')$. The restriction concerns their amenity values for that worker type. Experimentally constructed vacancies that differ in wages while holding other characteristics constant provide a potential source of such comparisons (Belot et al., 2022; Mas and Pallais, 2017).

Proposition 3.1 (Identification of the information cost)**.** *Suppose Assumption 3.1 holds. Fix a worker type x and a wage vector $\boldsymbol{w}$. Let y, y' be twins for that type, with $w(y) > w(y')$, and suppose they offer equal job-finding probabilities, $\rho_y^x(\boldsymbol{w}) = \rho_{y'}^x(\boldsymbol{w}) =: \rho$. Then*

$$k_x = \frac{\rho\big(w(y) - w(y')\big)}{\log \dfrac{\mu(y \mid x, \boldsymbol{w})}{\mu(y' \mid x, \boldsymbol{w})} - \log \dfrac{m(y \mid x)}{m(y' \mid x)}}, \tag{3.2}$$

provided the denominator is nonzero.

Proof. The twin restriction and equal job-finding probabilities imply

$$v_y^x(\boldsymbol{w}) - v_{y'}^x(\boldsymbol{w}) = \rho\big[\alpha(x, y) + w(y) - \alpha(x, y') - w(y')\big] = \rho\big(w(y) - w(y')\big).$$

Substituting into (3.1) gives (3.2). □

The numerator is the twins' difference in expected wage income. The denominator is the difference between their conditional and ex-ante log choice odds. Equal job-finding probabilities remove the effect of the common amenity value on the payoff difference, so that value can remain unknown. The formula becomes sensitive to errors in estimated shares when the denominator is close to zero.

3.1.1. Recovering the common amenity value

To recover the common amenity value, suppose the choice set contains a reference option, indexed by 1, with $v_1^x(\boldsymbol{w}) = 0$ for the worker type and wage states under consideration. If the choice set includes an option not to apply, it has zero payoff under the model's normalization.

An employed reference job with positive job-finding probability must satisfy $\alpha(x,1) = -w(1)$. Comparing a job with the reference option in (3.1) gives

$$\log \frac{\mu(y \mid x, \boldsymbol{w})}{\mu(1 \mid x, \boldsymbol{w})} - \log \frac{m(y \mid x)}{m(1 \mid x)} = \frac{\rho_y^x(\boldsymbol{w})\big(\alpha(x,y) + w(y)\big)}{k_x}. \tag{3.3}$$

Proposition 3.2 (Joint identification with a reference option)**.** *Suppose Assumption 3.1 holds. Fix x and a wage state with $v_1^x(\boldsymbol{w}) = 0$, and let y, y' be twins with $w(y) \neq w(y')$. Suppress the state argument of the job-finding probabilities and define*

$$a_1 = \log \frac{\mu(y \mid x, \boldsymbol{w})}{\mu(1 \mid x, \boldsymbol{w})} - \log \frac{m(y \mid x)}{m(1 \mid x)},$$
$$a_2 = \log \frac{\mu(y' \mid x, \boldsymbol{w})}{\mu(1 \mid x, \boldsymbol{w})} - \log \frac{m(y' \mid x)}{m(1 \mid x)}.$$

If

$$D := a_2 \rho_y^x - a_1 \rho_{y'}^x \neq 0, \tag{3.4}$$

then k_x and the common amenity value $\alpha = \alpha(x,y) = \alpha(x,y')$ are identified by

$$k_x = \frac{\rho_y^x \rho_{y'}^x \big(w(y') - w(y)\big)}{D}, \qquad \alpha = \frac{a_1 \rho_{y'}^x w(y') - a_2 \rho_y^x w(y)}{D}. \tag{3.5}$$

Proof. Equation (3.3) and the twin restriction give the linear system

$$a_1 k_x - \rho_y^x \alpha = \rho_y^x w(y), \qquad a_2 k_x - \rho_{y'}^x \alpha = \rho_{y'}^x w(y').$$

Its determinant is D. Solving the system yields (3.5). □

The zero-payoff reference option makes the common amenity value identifiable even when the twins have different job-finding probabilities. Under the model, $D = \rho_y^x \rho_{y'}^x (w(y') - w(y))/k_x$. Hence distinct wages and positive job-finding probabilities give a nonzero determinant for a finite information cost.

4. Estimation

The estimators initially take job-finding probabilities as given. Full-equilibrium estimation also imposes the matching equations. The generalized logit formula gives conditional application probabilities, and the outer information problem restricts their ex-ante weights. Those weights can be estimated before maximizing the likelihood or chosen jointly with the parameters subject to the optimality conditions.

4.1. The estimation problem

Suppose applications are observed in M markets indexed by t, with n_t sampled workers of a given type per market. For exposition, fix that worker type and suppress x. Let Y_{it} denote worker i's application choice in market t. The sampled markets represent wage realizations from the common prior G. For the likelihood below, sampled workers' applications are independent conditional on the wage vector and equilibrium profile.

First treat job-finding probabilities as known or fixed at estimates obtained by evaluating r_y^x at conditional application shares for all worker types. Let $\theta \in \Theta$ contain the amenity and information-cost parameters, and define the normalized application payoff:[1]

$$v_{yt}(\theta) = \frac{\rho_{yt}\big(\alpha_\theta(y) + w_t(y)\big)}{k(\theta)}.$$

The information cost remains a parameter within θ; Section 3 gives sufficient conditions for identifying it in wage units. Write $F_v(\cdot;\theta)$ for the distribution of the normalized payoff vector $\boldsymbol{v}_t(\theta)$. Given ex-ante weights $q \in \Delta(\mathcal{Y})$, the conditional probability is

$$P_{yt}(\theta, q) = \frac{q_y \exp\{v_{yt}(\theta)\}}{\sum_{j\in\mathcal{Y}} q_j \exp\{v_{jt}(\theta)\}}. \tag{4.1}$$

The probabilities depend only on differences between normalized payoff indices, so an unrestricted index specification requires a location normalization. The indices can be expressed relative to option 1. Assigning a particular job zero payoff is a separate restriction on amenities and wages, as in Section 3.1.

The conditional log-likelihood, or plug-in criterion when matching probabilities are estimated, is

$$\ell(\theta, q) = \sum_{t=1}^{M}\sum_{i=1}^{n_t}\sum_{y\in\mathcal{Y}} \mathbf{1}\{Y_{it} = y\} \log P_{yt}(\theta, q). \tag{4.2}$$

The optimal weights satisfy the averaging condition

$$q_y = \int \frac{q_y \exp(v_y)}{\sum_j q_j \exp(v_j)}\, dF_v(\boldsymbol{v};\theta), \qquad y \in \mathcal{Y}. \tag{4.3}$$

For positive weights, this averaging condition is the first-order condition of the outer information problem. At zero weights, the optimality inequalities in Section 4.3 must also be imposed. The averaging condition links conditional probabilities across markets through their common ex-ante weights.

[1] With a free payoff difference for every option in every market, the conditional logit fits positive shares exactly and the update in (4.4) leaves the initial weights unchanged.

4.2. Plug-in and nested pseudo-likelihood estimation

If the sampled markets are draws from the wage-state distribution, an initial estimate of the ex-ante shares is

$$\widehat{q}_y^{(0)} = \frac{1}{M}\sum_{t=1}^{M}\widehat{\mu}_{yt}, \qquad \widehat{\mu}_{yt} = \frac{1}{n_t}\sum_{i=1}^{n_t}\mathbf{1}\{Y_{it} = y\}.$$

Each market receives equal weight because the target is an average over wage states; pooling all workers would give more weight to markets with larger samples n_t. If the sampling design overrepresents some wage states, the market weights must instead recover their probabilities in the target distribution. The plug-in estimator maximizes $\ell(\theta, \widehat{q}^{(0)})$ over θ. Replacing unknown choice probabilities with sample estimates follows the logic of conditional-choice-probability methods (Hotz and Miller, 1993).

The model can then update the weights by averaging its predicted application probabilities across markets. Let $\widehat{\theta}^{(j)} \in \arg\max_\theta \ell(\theta, \widehat{q}^{(j)})$ and set

$$\widehat{q}_y^{(j+1)} = \frac{1}{M}\sum_{t=1}^{M} P_{yt}\big(\widehat{\theta}^{(j)}, \widehat{q}^{(j)}\big). \tag{4.4}$$

Re-estimating θ at the updated weights gives a nested pseudo-likelihood iteration of the type studied by Aguirregabiria and Mira (2002, 2007). Convergence depends on the specification and the update map (Kasahara and Shimotsu, 2012). For iteration, replace zero sample shares by positive starting values and renormalize the weights: the update cannot revive a zero weight. Any boundary solution must also satisfy the optimality inequalities for excluded options.

4.3. Constrained optimization

The constrained estimator chooses amenity values, information costs, and ex-ante weights jointly. Following Su and Judd (2012), it imposes the model's optimality conditions as constraints on likelihood maximization. For a specified or estimated payoff distribution $F_v(\cdot;\theta)$, define

$$D_y(\theta, q) = \int \frac{\exp(v_y)}{\sum_j q_j \exp(v_j)}\, dF_v(\boldsymbol{v};\theta) \in [0, +\infty].$$

The estimator solves

$$\begin{aligned} \max_{\theta\in\Theta,\, q\in\Delta(\mathcal{Y})} \quad & \ell(\theta, q) \\ \text{subject to} \quad & D_y(\theta, q) \le 1, \qquad y \in \mathcal{Y}, \\ & D_y(\theta, q) = 1 \quad \text{whenever } q_y > 0. \end{aligned} \tag{4.5}$$

These conditions impose equality for every option with positive weight and retain the inequality for unchosen options. They are necessary and sufficient for optimal weights; Appendix B.1 gives

the argument for general payoff distributions. This formulation holds matching probabilities fixed.

For the full RISE model, the optimization must also choose conditional application profiles for all worker types and evaluate payoffs at the matching probabilities they generate. The choice equations (2.10) and matching equations (2.12), together with the weight optimality conditions, must hold over the support of the wage distribution. The likelihood sums over all worker types.

4.4. Other computational approaches

The information problem can also be solved within a nested optimization routine using the Blahut–Arimoto algorithm (Blahut, 1972; Arimoto, 1972), with an outer likelihood step as in Rust (1987). Distributional restrictions offer another route. For example, Brown and Jeon (2024) develop a specification with tractable conditional choice probabilities under a Cardell prior.

Inference must account for estimated shares and any estimated inputs to the matching probabilities or wage distribution.

5. Model Comparison and Empirical Motivation

Information acquisition can change how applications respond to an expanded choice set and can produce zero application shares.

5.1. Monotonicity

Let $P_y(\mathcal{C})$ be the probability of choosing option y from a set $\mathcal{C}$. The monotonicity property considered here, also called regularity, is

$$P_y(\mathcal{C} \cup \{\tilde{y}\}) \leq P_y(\mathcal{C}), \qquad y \in \mathcal{C}.$$

Adding an option weakly reduces the probability of choosing each existing option.

5.1.1. Monotonicity in random utility models

In a random utility model, a worker chooses the option with the highest utility (Luce, 1959; McFadden, 1973). Fix a worker type x and let $(u_y^x)_y$ be a random utility vector whose joint distribution for existing options is unchanged when another option becomes available. Suppose the maximizer is unique almost surely, or that ties are resolved by a fixed priority rule.

The event that y is chosen from the expanded set is contained in the event that it is chosen from the original set, so $P_y(\mathcal{C} \cup \{\tilde{y}\}) \leq P_y(\mathcal{C})$.

5.1.2. *Monotonicity in directed search benchmarks*

In the symmetric single-application urn-ball environment with identical jobs, a worker applies to each of J jobs with probability $1/J$ (Peters, 1984, 1991). Adding an identical job lowers this probability to $1/(J+1)$. In the multiple-application environment of Belot et al. (2022), based on Albrecht et al. (2006), a worker sends N applications among L observed vacancies. If one vacancy has application probability p and the remaining vacancies are symmetric, each of the latter receives an application with probability $(N-p)/(L-1)$. Relative to N applications among $L-1$ symmetric jobs, the additional vacancy absorbs p applications in expectation.

5.1.3. *Non-monotonicity in rational inattention models*

Under rational inattention, adding an option can change the information a worker chooses to acquire. For a choice set $\mathcal{C}$, let $F^x_{\mathcal{C}}$ be an optimal information structure. After receiving signal $\boldsymbol{s}_i$, worker i chooses the option with the highest posterior expected payoff,

$$u^x_y(i;\mathcal{C}) = \int v^x_y(\boldsymbol{w};\mathcal{C})\, dF^x_{\mathcal{C}}(\boldsymbol{w} \mid \boldsymbol{s}_i), \qquad y \in \mathcal{C}. \tag{5.1}$$

Changing $\mathcal{C}$ can change these posterior expected payoffs through the optimal information structure and, in a search market, through congestion. The invariance condition used in Section 5.1.1 then need not hold.

Matějka and McKay (2015) and Liao (2024) give examples in which the probability of choosing an existing option rises after another option is added. The mechanism is a change in the return to learning. A new option that performs particularly well in one state can make information about that state more valuable. The resulting information can also help the worker recognize when an existing option is attractive.

5.1.4. *Occupational choices in the job-search data*

The platform studied by Belot et al. (2022) records which vacancies job seekers viewed and saved. Clicking a search result reveals the vacancy's full description, and saving a vacancy records an intention to consider it for application. I assign each subject to the occupation category in which she saved the most vacancies and calculate the share assigned to each category within groups defined by the categories viewed. Figure 2 compares subjects who viewed vacancies in all nine one-digit Standard Occupational Classification (SOC) categories with subjects who did not view Associate Professional and Technical occupations (SOC 3).

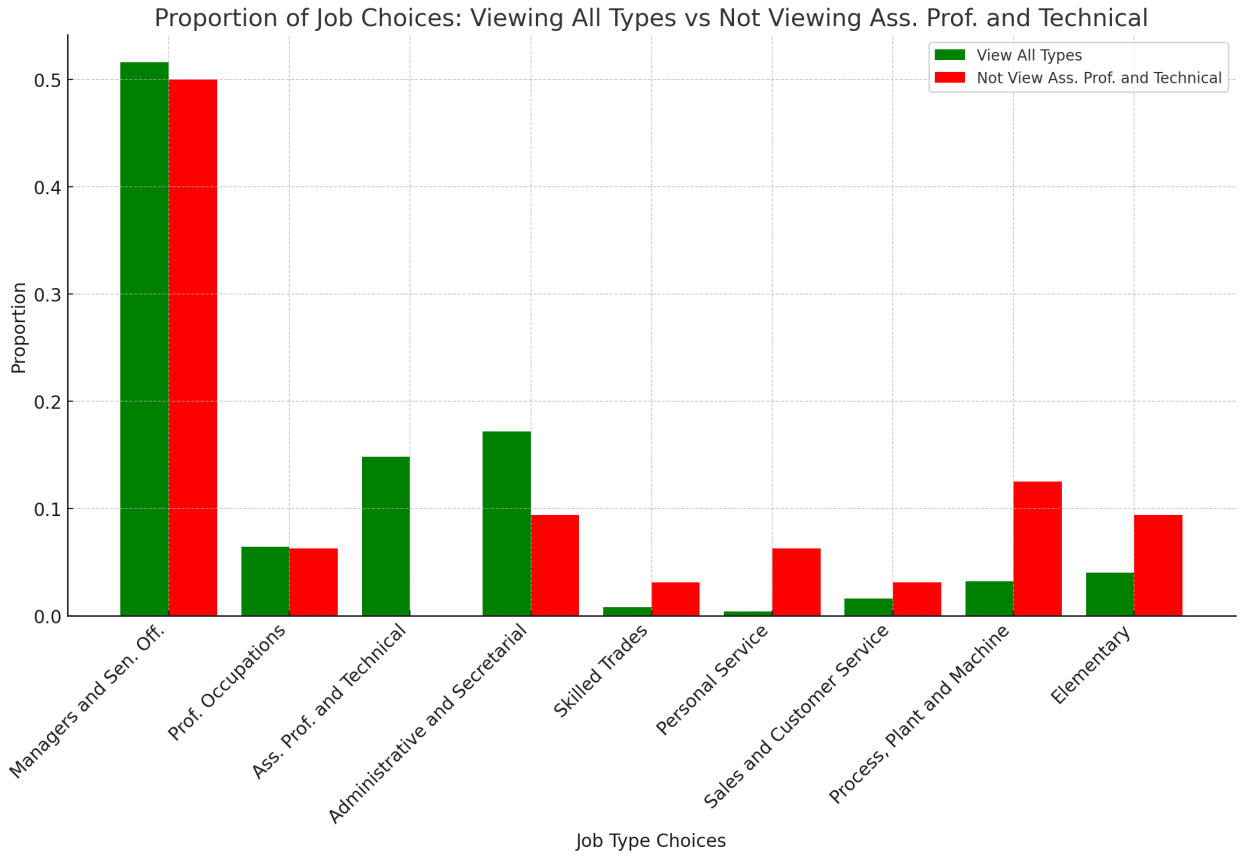


Figure 2: Occupational choice shares by categories viewed. Green bars show subjects who viewed all nine SOC categories; red bars show subjects who did not view SOC 3. A subject's choice is the category in which she saved the most vacancies.

Notes: SOC denotes Standard Occupational Classification. *Source:* Author's calculations using vacancy-viewing and saving records from Belot et al. (2022).

The share assigned to Administrative and Secretarial occupations (SOC 4) is higher among subjects who viewed all categories. This pattern motivates a model in which the set of options considered affects information acquisition and hence choices among existing options. Under rational inattention, learning about one occupation can also convey information about another when their payoffs are related in workers' beliefs. The groups in the figure are formed from subjects' viewing decisions, so differences in preferences and search behavior can also contribute to the observed shares.

5.2. Zero shares and kinks

Rational inattention also permits exact zero choice shares and kinks in choice probabilities (Matějka and McKay, 2015; Caplin et al., 2019). An option can receive zero ex-ante weight when the expected gain from considering it is too small to justify the information cost; equation (2.7) then assigns it zero conditional probability at every state. Changes in the set of options with

positive weights can produce kinks as payoffs vary. By comparison, additive random utility models with a joint shock distribution of full support assign every option positive probability, and the usual differentiable generalized extreme-value specifications have smooth choice probabilities (McFadden, 1978).

6. Conclusion

The model provides an equilibrium foundation for structural estimation of job search with heterogeneous workers and firms. The linear-fractional structure of the generalized logit formula supports existence despite multiple optimal information choices. Under the stated twin-job restrictions, application shares identify information costs. A zero-payoff reference option also permits recovery of the twins' common amenity value.

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A. Notation

The table collects the main objects used in the model, identification analysis, and estimation discussion.

Symbol	Meaning
$x \in \mathcal{X}$, $y \in \mathcal{Y}$	Worker type and firm or job type
$m_{\mathcal{J}}$	Mass of firms; the mass of workers is one
$\boldsymbol{w}$, $w(y)$	Wage vector and wage of job type y
G, $\mathcal{W}$	Prior over wage vectors and its support
$\alpha(x, y)$	Amenity value for worker type x, relative to unemployment
k_x	Unit cost of information for worker type x
$\rho_y^x(\boldsymbol{w})$	Job-finding probability after applying to y
r_y^x	Function mapping application probabilities to job-finding probabilities
$v_y^x(\boldsymbol{w})$	Expected application payoff, $\rho_y^x(\boldsymbol{w})(\alpha(x, y) + w(y))$
F	Joint distribution of signals and wage states
$\beta_y^x(\boldsymbol{w})$	Conditional application probability
$\bar{\beta}_y^x$	Ex-ante application probability
$\mu(y \mid x, \boldsymbol{w})$	Conditional share, equal to $\beta_y^{*x}(\boldsymbol{w})$
$m(y \mid x)$	Ex-ante share, equal to $\bar{\beta}_y^{*x}$
q	Candidate vector of ex-ante weights in the information problem
t, n_t	Market index and number of sampled workers in market t
Y_{it}	Application choice of worker i in market t
$F_v(\cdot; \theta)$	Distribution of normalized payoff vectors at parameter θ

B. Existence of RISE

This appendix proves Proposition 2.3. The proof allows the worker's outer problem to have multiple maximizers and includes the case of zero ex-ante shares.

Proof. Write $\mathcal{W} = \operatorname{supp}(G)$ and $p_{\boldsymbol{w}} = G(\{\boldsymbol{w}\}) > 0$. Let

$$\Delta = \left\{ q \in \mathbb{R}_+^{|\mathcal{Y}|} : \sum_{y \in \mathcal{Y}} q_y = 1 \right\}, \qquad K = \prod_{x \in \mathcal{X},\, \boldsymbol{w} \in \mathcal{W}} \Delta.$$

Thus K is the nonempty, compact, convex set of state-contingent application profiles. I construct a correspondence $H : K \rightrightarrows K$ of optimal application profiles and show that its fixed points satisfy the equilibrium definition.

Step 1: Optimal ex-ante choice weights. For a conjectured profile $\boldsymbol{b} \in K$, define

$$g_y^x(\boldsymbol{w};\boldsymbol{b}) = \exp\left\{ \frac{r_y^x\big((b_y^{x'}(\boldsymbol{w}))_{x'\in\mathcal{X}}\big)\big(\alpha(x,y)+w(y)\big)}{k_x} \right\},$$

and, for $q \in \Delta$, let

$$Z_{\boldsymbol{w}}^x(q;\boldsymbol{b}) = \sum_{y\in\mathcal{Y}} q_y g_y^x(\boldsymbol{w};\boldsymbol{b}), \qquad U^x(q;\boldsymbol{b}) = k_x \sum_{\boldsymbol{w}\in\mathcal{W}} p_{\boldsymbol{w}} \log Z_{\boldsymbol{w}}^x(q;\boldsymbol{b}).$$

Each g_y^x is continuous and strictly positive. Since the index sets are finite and K is compact, the denominators $Z_{\boldsymbol{w}}^x$ are uniformly bounded away from zero on $\Delta \times K$. Consequently, U^x is jointly continuous and concave in q. Its maximizer correspondence

$$B^x(\boldsymbol{b}) = \arg\max_{q\in\Delta} U^x(q;\boldsymbol{b})$$

has nonempty compact values because Δ is compact and U^x is continuous. Concavity of U^x makes these maximizer sets convex. Since the feasible set Δ does not depend on $\boldsymbol{b}$, the maximum theorem gives upper hemicontinuity of B^x and a closed graph.

Step 2: All optimal weights give the same denominator at each state. Fix x and $\boldsymbol{b}$, and take $q, \tilde{q} \in B^x(\boldsymbol{b})$. The denominators are identical across these maximizers:

$$Z_{\boldsymbol{w}}^x(q;\boldsymbol{b}) = Z_{\boldsymbol{w}}^x(\tilde{q};\boldsymbol{b}) \qquad \text{for every } \boldsymbol{w} \in \mathcal{W}. \tag{B.1}$$

Indeed, $Z_{\boldsymbol{w}}^x$ is linear in q, so strict concavity of the logarithm gives

$$\log Z_{\boldsymbol{w}}^x\left(\frac{q+\tilde{q}}{2};\boldsymbol{b}\right) \geq \frac{1}{2}\log Z_{\boldsymbol{w}}^x(q;\boldsymbol{b}) + \frac{1}{2}\log Z_{\boldsymbol{w}}^x(\tilde{q};\boldsymbol{b}),$$

with strict inequality whenever the two denominators differ. Because $k_x > 0$ and every support state has positive probability, a difference at even one state would imply

$$U^x\left(\frac{q+\tilde{q}}{2};\boldsymbol{b}\right) > \frac{1}{2}U^x(q;\boldsymbol{b}) + \frac{1}{2}U^x(\tilde{q};\boldsymbol{b}).$$

The midpoint lies in Δ, so this inequality contradicts optimality of q and $\tilde{q}$. This proves (B.1). The denominator may vary across states, but at each state it is constant over the entire optimizer set $B^x(\boldsymbol{b})$.

Step 3: The correspondence of conditional application profiles. Define

$$T_y^x(\boldsymbol{w};q,\boldsymbol{b}) = \frac{q_y g_y^x(\boldsymbol{w};\boldsymbol{b})}{Z_{\boldsymbol{w}}^x(q;\boldsymbol{b})},$$

and let

$$H(\boldsymbol{b}) = \left\{ \left(T_y^x(\boldsymbol{w}; q^x, \boldsymbol{b}) \right)_{x,y,\boldsymbol{w}} : q^x \in B^x(\boldsymbol{b}) \text{ for every } x \in \mathcal{X} \right\}.$$

For each worker type, a single weight vector q^x is used at every state. The components of T^x are nonnegative and sum to one at each state, so $H(\boldsymbol{b}) \subseteq K$. Joint continuity of T and compactness of the optimizer sets imply that H has nonempty compact values.

To establish convexity, fix $\boldsymbol{b}$, choose $q^x, \tilde{q}^x \in B^x(\boldsymbol{b})$ for each x, and let $q_\theta^x = \theta q^x + (1-\theta)\tilde{q}^x$, where $\theta \in [0,1]$. Convexity of $B^x(\boldsymbol{b})$ implies $q_\theta^x \in B^x(\boldsymbol{b})$. By (B.1),

$$T_y^x(\boldsymbol{w}; q_\theta^x, \boldsymbol{b}) = \theta T_y^x(\boldsymbol{w}; q^x, \boldsymbol{b}) + (1-\theta) T_y^x(\boldsymbol{w}; \tilde{q}^x, \boldsymbol{b})$$

simultaneously for every $x, y, \boldsymbol{w}$. Hence $H(\boldsymbol{b})$ is convex. Although T^x is generally linear-fractional in the weights at each state, it is affine on the optimizer set jointly across states.

The graph of H is closed. To see this, take $\boldsymbol{b}_n \to \boldsymbol{b}$ and $h_n \to h$ with $h_n \in H(\boldsymbol{b}_n)$, and choose representing weights $q_n^x \in B^x(\boldsymbol{b}_n)$. Compactness of $\prod_x \Delta$ yields a subsequence $(n_\ell)_\ell$ along which $q_{n_\ell}^x \to q^x$ for every x. The closed graph of B^x gives $q^x \in B^x(\boldsymbol{b})$. Continuity of T then implies

$$h = \lim_{\ell \to \infty} h_{n_\ell} = \left(T_y^x(\boldsymbol{w}; q^x, \boldsymbol{b}) \right)_{x,y,\boldsymbol{w}}.$$

Thus $h \in H(\boldsymbol{b})$. Since the range K is compact, the closed graph implies upper hemicontinuity of H.

Step 4: Consistency of ex-ante and conditional probabilities. For $q \in \Delta$, define the normalized derivatives

$$D_y^x(q; \boldsymbol{b}) = \frac{1}{k_x} \frac{\partial U^x(q; \boldsymbol{b})}{\partial q_y} = \sum_{\boldsymbol{w} \in \mathcal{W}} p_{\boldsymbol{w}} \frac{g_y^x(\boldsymbol{w}; \boldsymbol{b})}{Z_{\boldsymbol{w}}^x(q; \boldsymbol{b})}.$$

Their q-weighted average is

$$\sum_y q_y D_y^x(q; \boldsymbol{b}) = \sum_{\boldsymbol{w} \in \mathcal{W}} p_{\boldsymbol{w}} \frac{\sum_y q_y g_y^x(\boldsymbol{w}; \boldsymbol{b})}{Z_{\boldsymbol{w}}^x(q; \boldsymbol{b})} = 1.$$

Now take $q \in B^x(\boldsymbol{b})$, and let e_y denote the unit vector for option y. Optimality of q requires the directional derivative toward e_y to be nonpositive:

$$\frac{d}{dt} U^x\big((1-t)q + t e_y; \boldsymbol{b}\big)\Big|_{t=0^+} = k_x \big(D_y^x(q; \boldsymbol{b}) - 1 \big) \le 0.$$

Thus $D_y^x(q; \boldsymbol{b}) \le 1$ for all y. A strict inequality for an option with $q_y > 0$ would make their q-weighted average less than one. Hence $D_y^x(q; \boldsymbol{b}) = 1$ whenever $q_y > 0$, and

$$\sum_{\boldsymbol{w} \in \mathcal{W}} p_{\boldsymbol{w}} T_y^x(\boldsymbol{w}; q, \boldsymbol{b}) = q_y D_y^x(q; \boldsymbol{b}) = q_y. \tag{B.2}$$

This also holds when $q_y = 0$, since the induced conditional probability is then zero at every state. Therefore every optimal weight vector is exactly the ex-ante average of the conditional probabilities it generates.

Conversely, let $q \in \Delta$ satisfy $D_y^x(q;\boldsymbol{b}) \le 1$ and $q_y[D_y^x(q;\boldsymbol{b}) - 1] = 0$ for every y. Concavity of U^x gives, for every $q' \in \Delta$,

$$U^x(q';\boldsymbol{b}) - U^x(q;\boldsymbol{b}) \le k_x \sum_y D_y^x(q;\boldsymbol{b})(q'_y - q_y) \le 0.$$

Thus these conditions are sufficient as well as necessary for $q \in B^x(\boldsymbol{b})$.

Step 5: A fixed point is a RISE. The correspondence H maps the nonempty compact convex set K into itself, is upper hemicontinuous, and has nonempty compact convex values. Kakutani's fixed-point theorem therefore gives $\boldsymbol{\beta}^* \in H(\boldsymbol{\beta}^*)$. Choose representing weights $q^{*x} \in B^x(\boldsymbol{\beta}^*)$. Equation (B.2) implies

$$q_y^{*x} = \sum_{\boldsymbol{w} \in \mathcal{W}} p_{\boldsymbol{w}} \beta_y^{*x}(\boldsymbol{w}) = \bar{\beta}_y^{*x}.$$

The fixed-point relation gives the required conditional choice formula; $q^{*x} \in B^x(\boldsymbol{\beta}^*)$ gives optimality of the ex-ante weights; and the definition of g_y^x evaluates job-finding probabilities at the same profile $\boldsymbol{\beta}^*$. These are all three conditions in the definition of RISE. □

B.1. Weight optimality for general payoff distributions

Fix θ and let $\mathbb{E}_\theta$ denote expectation under $F_v(\cdot;\theta)$. The model's finite wage moments and $k(\theta) > 0$ imply $\mathbb{E}_\theta \max_y |v_y| < \infty$. For $q \in \Delta(\mathcal{Y})$, write

$$Z(\boldsymbol{v}, q) = \sum_j q_j e^{v_j}, \qquad U_\theta(q) = \mathbb{E}_\theta \log Z(\boldsymbol{v}, q).$$

The objective is finite because $\min_j v_j \le \log Z(\boldsymbol{v}, q) \le \max_j v_j$. For option y, let e_y be its unit vector and set $a_y(\boldsymbol{v}, q) = e^{v_y}/Z(\boldsymbol{v}, q) - 1 \ge -1$. As $t \downarrow 0$, the quotients $\log(1 + t a_y)/t$ increase to a_y and are bounded below by -2 for $0 < t \le 1/2$. Monotone convergence therefore gives

$$\lim_{t \downarrow 0} \frac{U_\theta((1-t)q + t e_y) - U_\theta(q)}{t} = \mathbb{E}_\theta a_y(\boldsymbol{v}, q) = D_y(\theta, q) - 1,$$

where the limit may be $+\infty$. At a maximizer these limits are nonpositive, so every $D_y(\theta, q) \le 1$. The identity $\sum_y q_y D_y(\theta, q) = 1$ then gives equality whenever $q_y > 0$. Conversely, a vector q satisfying the conditions in (4.5) maximizes U_θ, since for every $q' \in \Delta(\mathcal{Y})$, $\log z \le z - 1$ gives

$$U_\theta(q') - U_\theta(q) = \mathbb{E}_\theta \log \frac{Z(\boldsymbol{v}, q')}{Z(\boldsymbol{v}, q)} \le \sum_y q'_y D_y(\theta, q) - 1 \le 0.$$

C. An Equivalent Formulation of RISE

The worker optimality conditions in Definition 2.3 can be stated directly as an optimization over recommendation strategies. For a given application profile $\boldsymbol{\beta}^*$, define

$$\rho_y^{*x}(\boldsymbol{w}) = r_y^x\big(\beta_y^{*1}(\boldsymbol{w}), \ldots, \beta_y^{*|\mathcal{X}|}(\boldsymbol{w})\big).$$

The profile is a RISE if and only if, for every worker type x,

$$\boldsymbol{\beta}^{*x} \in \arg\max_{\boldsymbol{\beta}} \Big\{ \int \sum_y \rho_y^{*x}(\boldsymbol{w})\big(\alpha(x,y) + w(y)\big)\beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}) - c_x(\boldsymbol{\beta}, G) \Big\}, \tag{C.1}$$

where the maximization is over recommendation strategies and

$$c_x(\boldsymbol{\beta}, G) = k_x \left[\int \sum_y \beta_y(\boldsymbol{w}) \log \beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}) - \sum_y \bar{\beta}_y \log \bar{\beta}_y \right], \qquad \bar{\beta}_y = \int \beta_y(\boldsymbol{w})\, dG(\boldsymbol{w}).$$

Proposition 2.2 gives the equivalence between (C.1) and conditions (i)–(ii) of Definition 2.3. In both formulations, a worker takes the job-finding probabilities generated by the population profile as given.

D. A Two-Type Matching Example

Consider two job types, each with one unit of vacancies, and two worker types $x \in \{A, B\}$, each of mass N. The worker types differ only in their information costs $k_A, k_B > 0$, and amenity values are zero. Wages take two possible values:

	State 1	State 2
$w(1)$	0	1
$w(2)$	R	R
Probability	g_0	$1-g_0$

$$0 < R < 1, \quad 0 < g_0 < 1.$$

This example uses worker mass $2N$ to vary congestion relative to the two units of vacancies. Write the wage states as $\boldsymbol{w}_1 = (0, R)$ and $\boldsymbol{w}_2 = (1, R)$, with probabilities $p_1 = g_0$ and $p_2 = 1-g_0$. Write $\beta^x_{ys} = \beta^x_y(\boldsymbol{w}_s)$ for the application probability in state s. Under urn-ball matching, the expected queue length in submarket y is

$$\lambda_{ys} = N\left(\beta^A_{ys} + \beta^B_{ys}\right),$$

and the job-finding probability is

$$\rho(\lambda) = \begin{cases} (1-e^{-\lambda})/\lambda, & \lambda > 0, \\ 1, & \lambda = 0. \end{cases}$$

The value at zero is the continuous extension. A worker who applies to an empty submarket is hired with probability one.

Let q^x_y denote type x's ex-ante application weight and define $v_{ys} = \rho(\lambda_{ys}) w_s(y)$. The equilibrium conditions can be written compactly as

$$\beta^x_{ys} = \frac{q^x_y \exp(v_{ys}/k_x)}{\sum_{j=1}^2 q^x_j \exp(v_{js}/k_x)}, \qquad x \in \{A, B\},\ y, s \in \{1, 2\}, \tag{D.1}$$

$$q^x_y = \sum_{s=1}^2 p_s \beta^x_{ys}, \qquad q^x_1 + q^x_2 = 1, \quad q^x_y \geq 0. \tag{D.2}$$

The ex-ante weights must also solve the outer information problem. Equivalently, they satisfy

$$\sum_{s=1}^2 p_s \frac{\exp(v_{ys}/k_x)}{\sum_{j=1}^2 q^x_j \exp(v_{js}/k_x)} \begin{cases} = 1, & q^x_y > 0, \\ \leq 1, & q^x_y = 0. \end{cases}$$

After rescaling worker and vacancy masses to the main text's normalization, Proposition 2.3 ensures equilibrium existence.